

Booking lifecycle and contention

From trader intent through to authoritative inventory allocation

I chose to go deepest on **booking lifecycle and contention**.

The architectural principle is:

Commercial intent is optimistic; inventory allocation is pessimistic.

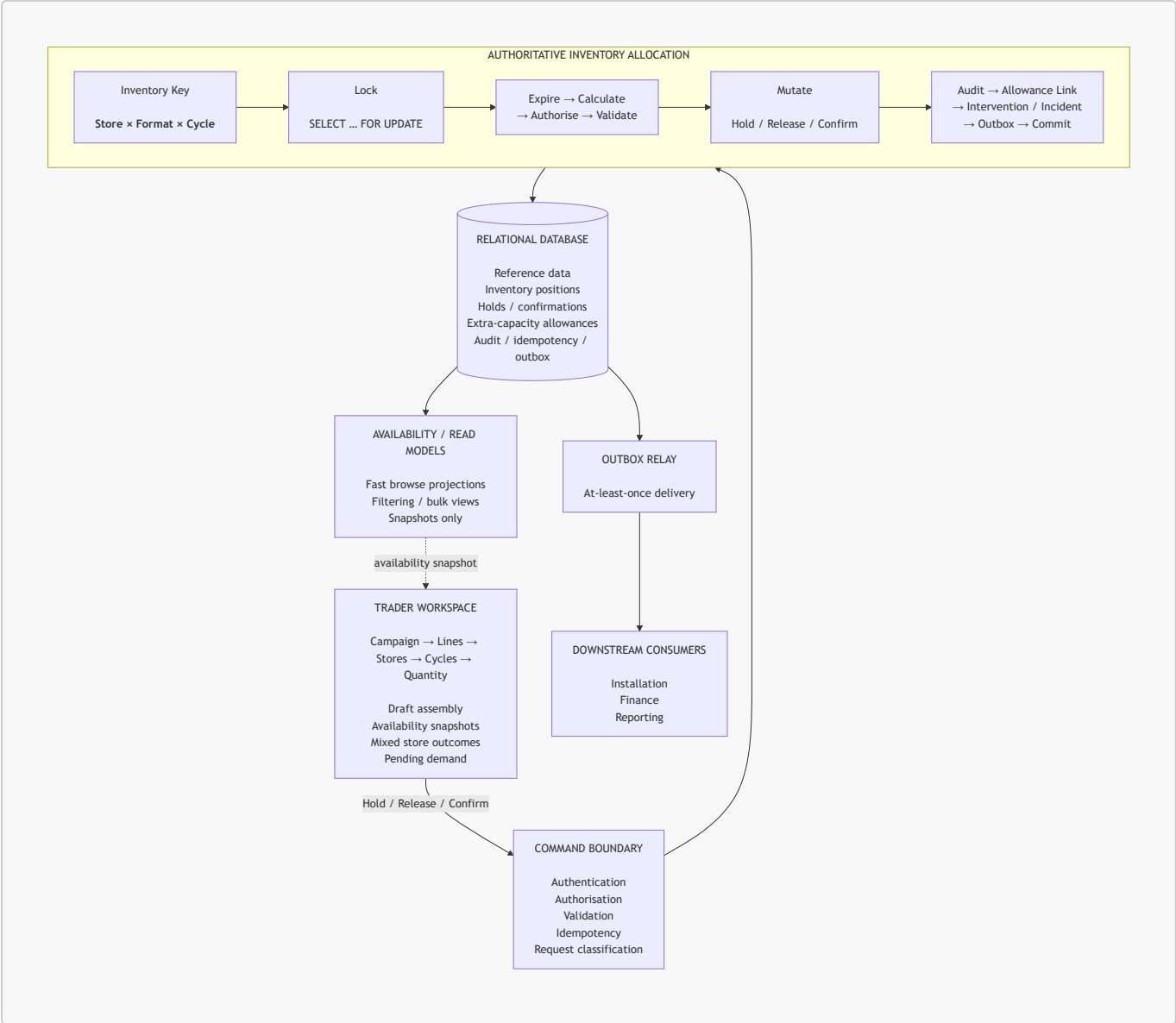
Traders should be able to assemble and revise campaign demand freely. But the moment an action consumes contested physical inventory — placing a hold or confirming an allocation — it crosses a strongly consistent transactional boundary.

That is the part of the system I would protect most carefully.

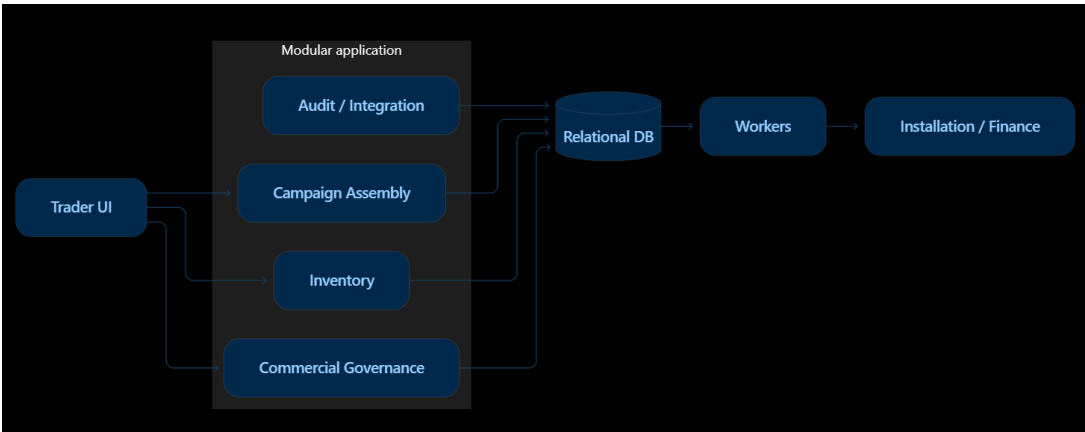
If availability search is slow, traders are frustrated. If allocation is wrong, two advertisers can both be told they successfully acquired the same finite physical inventory. That failure propagates into campaign operations, installation and downstream commercial data.

This is therefore a **two-hour depth slice**, not an attempt to specify the entire platform.

1. What does the system look like end-to-end?



I would start with a **modular application backed by PostgreSQL as the authoritative system of record.**



High Level System boundaries and deployment

The system naturally separates into four concerns:

Area	Responsibility	Consistency
Reference data	stores, formats, physical capacity, trading cycles	read-heavy, relatively static

Area	Responsibility	Consistency
Campaign assembly	advertisers, campaigns, lines, store selections, requested quantities, pending demand	optimistic
Inventory truth	holds, confirmations, releases, expiry, exceptional-capacity authority	strong transactional
Audit & integration	material transitions, actors, idempotency, downstream publication	durable

These are consistency and ownership boundaries, not a requirement for four services.

A modular monolith is sufficient initially.

Availability is computed, then immediately untrusted

The natural contention key is:

```
(store, format, cycle)
```

For example:

```
Store 142
Aisle Barrier
Cycle 8
```

A trader might see:

```
Physical capacity    6
Confirmed            3
Active holds         2
Available            1
```

But **Available = 1** is only true at the instant it was calculated. Another trader may consume it immediately afterwards.

The UI, browse query and any cache can tell a trader what appeared available. Only the allocation transaction can tell them what they actually acquired.

Demand is not inventory

A campaign line expresses demand.

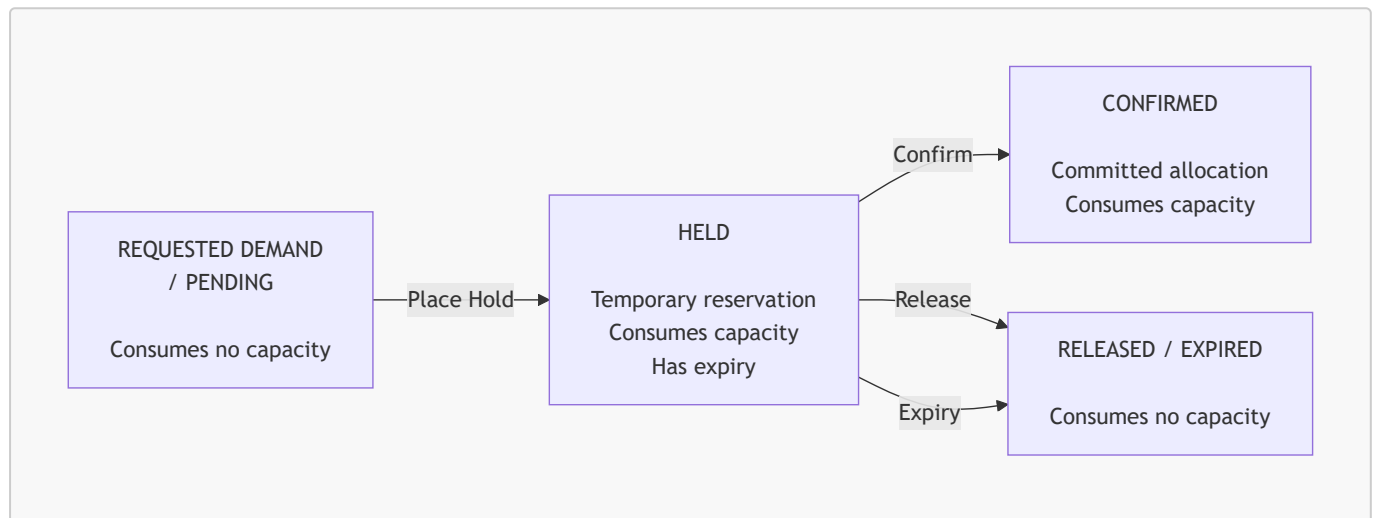
For example:

Campaign line

Format	Aisle Barrier
Store	142
Cycle	8
Requested	4

That request consumes nothing simply by existing.

The inventory lifecycle is separate.

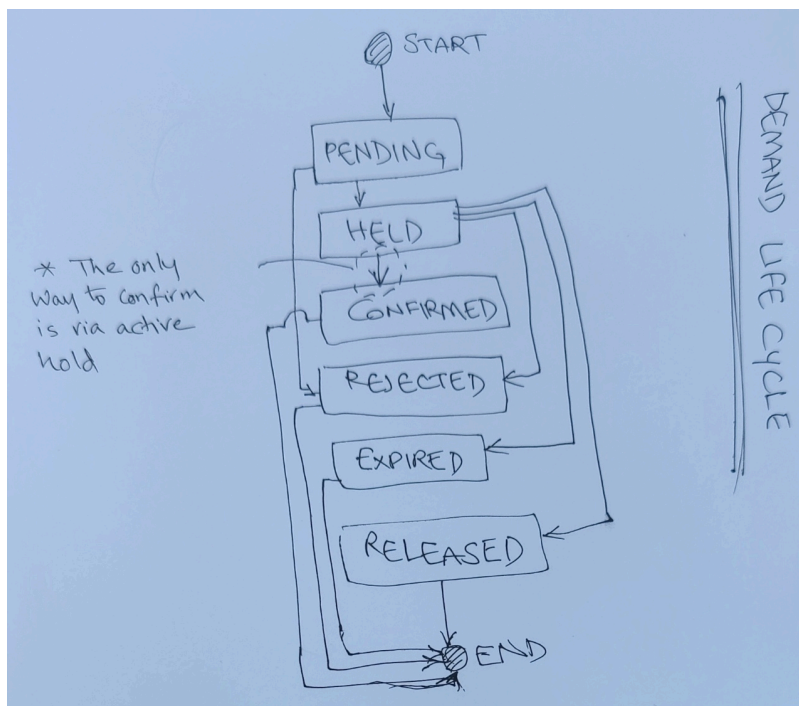


There is also **PENDING AVAILABILITY** which means *The advertiser still wants this quantity, but no capacity has been reserved.*

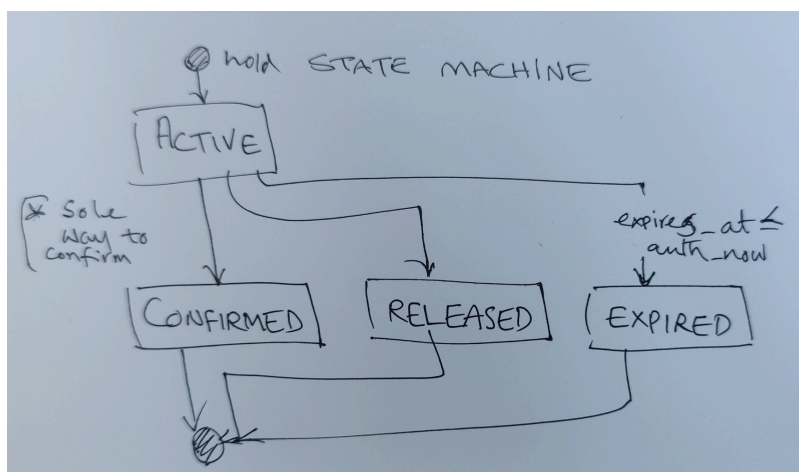
Pending demand:

- consumes no capacity;
- carries no guarantee;
- establishes no priority;
- is not automatically allocated if capacity later becomes free.

That distinction prevents commercial intent from accidentally becoming inventory truth.



A demand (full) life cycle - The sole way to confirm is via an ACTIVE HOLD



HOLD state machine diagram

Pending is a state to help Traders create their campaign before commit to any booking, including **Hold** (provisional/temp booking), or while awaiting for a **Hold** (such as in case of awaiting an **Exception**, etc.).

Quantity is (strictly) conserved

For each demand item, quantity can move between dispositions, but must never disappear or be duplicated.

Suppose the requested quantity is 4 and only two are currently bookable. After the advertiser accepts that partial outcome:

Held	2
Pending availability	2
Confirmed	0
Rejected	0

Total	4
-------	---

The invariant is:

Every unit of requested demand has exactly one current disposition.

A transition may move quantity between states.

It may not silently create, duplicate or lose quantity (no accidental shrinkage or growth).

2. Where did I choose to go deep, and why?

I chose to go deepest on the transition from:

"I would like this inventory"

to:

"The business has now reserved or committed this inventory."

More specifically:

How does the system guarantee that concurrent legitimate actions cannot accidentally consume the same finite capacity?

That I'd say is the highest-risk engineering boundary in this problem.

The core invariant is:

No successful allocation command may consume capacity that was neither physically available nor covered by valid prior commercial authority.

The authoritative inventory position

I would maintain one `inventory_position` for each `(store_id, format_id, cycle_id)`.

Conceptually:

```
// inventory_position

store_id
format_id
```

```
cycle_id

active_held_quantity
confirmed_quantity

updated_at
```

That row has two jobs:

1. provide an efficient current aggregate;
2. provide the lock target for competing allocation transactions.

I would still retain individual allocation records.

So the design is deliberately hybrid:

```
Allocation records
  = explainable business facts

InventoryPosition
  = current aggregate + concurrency lock

Audit history
  = immutable material transition history
```

I would **not** store a naked `available_quantity` as independent truth. Because *Availability is derived*.

Hold and Confirm answer different questions

A useful distinction is:

HOLD: *"May I reserve this capacity now?"*

versus:

CONFIRM: *"Is my reservation still valid, and may it now become committed?"*

Hold

Hold is where new capacity is normally acquired.

It therefore carries the primary last-unit contention problem.

Confirm

Confirmation may not increase total consumed quantity: (because it's usually converts a hold)

```
held      -1
confirmed +1
```

but it still changes shared authoritative state.

It must therefore use the same inventory serialisation boundary.

It also needs to establish:

- that the hold still exists;
- that it has not expired;
- that it has not already been released or confirmed;
- that any exceptional-capacity authority is valid for the **resulting confirmed disposition**.

So **Confirm** is shorter than first-touch allocation.

But, it is not lock-free.

Concrete transaction choice

For this design I would use:

```
PostgreSQL
READ COMMITTED
+
explicit SELECT ... FOR UPDATE
```

on the affected **inventory_position**.

SERIALIZABLE would also be credible.

I prefer explicit row locking here because:

- contention is *expected* rather than unusual;
- the number of concurrent traders is modest;
- critical sections should be short;
- the lock target maps directly to the business contention key;
- the resulting behaviour is straightforward to reason about and test.

The design does not rely on PostgreSQL-specific business semantics, but PostgreSQL gives a concrete implementation model for proving the races.

The Hold transaction

A Hold command is approximately:

BEGIN

1. Claim / validate idempotency key
2. Establish authoritative database time
3. Lock:
 inventory_position(store, format, cycle)
 FOR UPDATE
4. Materialise any holds already logically expired
 at that authoritative time

5. Calculate:
 consumed =
 confirmed_quantity
 + active_held_quantity

6. Read physical capacity

7. Determine applicable approved extra capacity,
 if this operation is eligible to use any

bookable =
 physical
+ applicable extra
- consumed

8. Validate:
 demand state
 requested quantity
 capacity
 allowance entitlement

9. If requested > bookable:
 write no allocation for this item
 create terminal idempotent result:
 outcome = capacity_conflict
 requested_quantity = requested
 offered_quantity = bookable
 evaluated_at = now
 return conflict + non-binding offer

10. If requested <= bookable (valid):
 create Hold
 update InventoryPosition
 record allowance coverage if required
 record audit evidence
 if resulting consumed > physical capacity:
 create/update authorised-overcapacity (commercial) intervention
 write outbox event

```

        create terminal idempotent result:
            outcome = held
            allocated_id = ...
COMMIT

```

The important point is:

Capacity validation and capacity consumption happen inside the same transaction while the contention key is locked.

The hot transaction does not scan years of allocation history.

The current aggregate answers the allocation question; the underlying allocation facts and reconciliation prove that aggregate remains trustworthy.

The idempotency point is:

A committed business outcome is idempotent whether it succeeded or legitimately failed. A rolled-back technical failure is not.

 Update:

If I have more time think I can tighten up this transaction

This will check the authority BEFORE allocation vs allocation and then check whether we can justify it afterwards

```

...
7. Calculate ordinary availability:
    consumed =
        confirmed_quantity
        + active_held_quantity

    ordinary_available =
        max(0, physical_capacity - consumed)

8. Determine whether exceptional capacity is required:

    exceptional_required =
        max(0, requested - ordinary_available)

9. If exceptional_required > 0:

    find applicable PRIOR-approved extra-capacity allowance

    lock allowance entitlement if required

    validate:
        correct Store × Format × Cycle scope
        eligible actor / campaign / action
        HOLD disposition permitted

```

```

    allowance still available for new use
    remaining allowance >= exceptional_required

if validation fails:
    write no allocation

    persist terminal idempotent result:
        outcome = capacity_conflict
        requested_quantity = requested
        offered_quantity =
            ordinary_available
            + valid_authorised_extra_available

    return conflict + non-binding offer

```

10. If the full requested quantity is now authorised:

```

    create Hold

    update InventoryPosition

    if exceptional_required > 0:
        record durable allowance coverage
        consume/reserve exceptional_required
        against that allowance

    record audit evidence

    if resulting consumed > physical_capacity:
        create/update authorised-overcapacity intervention

    write outbox event

    persist terminal idempotent result:
        outcome = held
        allocation_id = ...

```

Race 1 — two traders request the last unit

Initial state:

Physical capacity	6
Confirmed	3
Held	2
Available	1

Trader A and Trader B have both already seen:

Trader A

Available = 1

Trader B

Available = 1

Then:

Trader A

BEGIN

lock inventory row

recalculate available = 1

create hold
held = 3

COMMIT

Trader B

BEGIN

attempts same lock
waits

lock acquired

recalculate available = 0

cannot create hold

return conflict / unavailable

Both traders can race.

Only one can win.

That is the required behaviour.

Bulk operations and deadlocks

A campaign may contain hundreds or thousands of store requests.

I would **not** lock an entire campaign in one transaction.

Normal atomicity is per independently contested demand item:

store × format × cycle

Bulk Place Hold uses the requested quantity only. There is no open human decision in that command. If **requested** > **bookable**, do not write a smaller hold. Return a conflict and a non-binding offer.

So one bulk action may return:

```
Store 001      HELD 4
Store 002      HELD 4
Store 003      UNAVAILABLE      bookable = 0
Store 004      CONFLICT          bookable = 2, requested = 4
                                   no stock taken
                                   companion offer: OFFERED 2 / PENDING 2
...

```

If the trader later submits Accept Offer { **accepted: 2**, **pending: 2** } and bookable is still 2:

```
Store 004      HELD 2
                PENDING 2

```

If bookable has since fallen to 1:

```
Store 004      STALE_OFFER
                allocation mutation = none
                revised offer: OFFERED 1 / PENDING 3

```

One bulk Hold may succeed on some stores and fail on others. That is required. A store that cannot satisfy the requested quantity is not held "a bit". It is **UNAVAILABLE** or **CONFLICT** plus a non-binding offer. Quantity splits only after the advertiser's decision is recorded on the accept command.

That keeps mixed store outcomes and lock duration small, without silent quantity shrink. If a transaction genuinely needs more than one inventory row, locks should be acquired in deterministic order, for example:

```
store_id
→ format_id
→ cycle_sequence

```

That removes a major class of deadlocks.

Deadlock or serialisation failure must still be treated as a normal retrieable transaction failure rather than something assumed impossible.

Race 2 — expiry versus confirmation

Suppose:

```
expires_at = 10:00:00
```

The database clock is authoritative.

I would define the boundary explicitly:

```
expires_at > authoritative_now  
  → active  
  
expires_at <= authoritative_now  
  → expired
```

Therefore:

```
expires_at = 10:00:00  
now        = 10:00:00
```

means the hold has expired.

Confirmation cannot resurrect it.

Expiry must not depend on a worker

I would still run a background process (hygiene process) to:

- materialise expired status;
- publish notifications;
- tidy read models;
- perform housekeeping.

But that process is not part of inventory correctness.

If it is delayed for an hour, the authoritative transaction still evaluates:

```
expires_at <= database_now
```

and treats the hold as expired.

So a worker outage can delay presentation.

It cannot strand expired capacity.

Confirm transaction

For this depth slice I would make one reversible design choice:

Confirmation requires an active Hold.

That is not a requirement from the brief. It is a first-slice simplification.

It gives the model one path for first-touch inventory acquisition:

```
Demand
↓
Hold      ← creates the allocation and acquires capacity
↓
Confirm   ← updates that same allocation into a commitment
```

A future direct allocate-and-confirm command could reuse the same authoritative allocation boundary if traders genuinely need it. It is outside this slice.

One allocation identity

Hold creates one allocation record:

```
allocation

id
demand_item_id
inventory_position_id
quantity

status      = held
expires_at   = ...
confirmed_at = null
```

If exceptional capacity was required, the **Hold** transaction also records durable allowance coverage (ie. audit record) showing which prior authority permitted that capacity to be consumed.

Confirm updates the **same allocation row**:

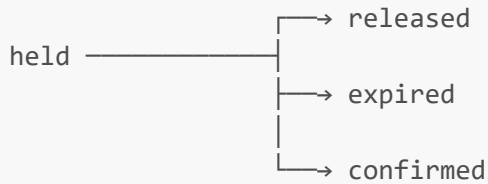
```
status      held → confirmed
confirmed_at = authoritative database now
```

- **expires_at** is kept as historical evidence of the original **Hold**. It governs the allocation only while **status = held**; once **Confirmed**, the allocation does not expire when the former **Hold** deadline

passes.

- While the allocation is held, manual **Release** or expiry transitions that same allocation row to released or expired; **Confirm** transitions it to confirmed.

Confirmed is terminal for this slice.



Material transitions are also written to append-only history:

```
HOLD_PLACED  
HOLD_RELEASED  
HOLD_EXPIRED  
HOLD_CONFIRMED
```

This gives the system:

```
one current allocation identity  
+  
an immutable history of how it changed
```

For this slice, **Confirm** transitions the entire held quantity. If partial confirmation is later required, the held quantity must be split explicitly. For example into a confirmed portion and a remaining held or released portion, so quantity is conserved.

Lock order

All inventory-mutating commands locks in the same order:

1. `inventory_position` rows
ordered by:
(`store_id`, `format_id`, `cycle_sequence`)
2. `allocation / hold` rows
ordered by `allocation_id`
3. `extra_capacity_allowance` rows
ordered by `allowance_id`
when new or changed allowance entitlement is required

Confirm locks the position before the hold row. **Confirm** does NOT take new extra capacity, so it does not need the allowance lock.

A consistent global order removes a major class of deadlocks. Deadlock or serialisation failure is still treated as a retrievable transaction outcome rather than assumed impossible.

Before I turn this **Hold** into a commitment, first make sure this isn't a retry. Use the database's clock. Lock the inventory key and the Hold so nobody can change the relevant state underneath me. If the Hold has already expired, materialise that expiry and refuse confirmation. If the Hold relied on exceptional capacity, prove that its prior authority allows that capacity to become Confirmed. Then transition the same allocation from Held to Confirmed, move the quantity between the held and confirmed counters, surface any authorised physical fulfilment risk, prove there is no unexplained overallocation, record the downstream event and idempotent result, and commit it atomically.

The transaction

BEGIN

1. Idempotency key
 same key + same fingerprint → stored result
 same key + different body → reject

2. Authoritative now = database time

3. Lock inventory_position FOR UPDATE
 then lock the allocation row

4. Require:
 allocation exists
 status = held
 quantity is this hold's quantity

Evaluate expiry using authoritative database time:

```
if expires_at <= now:
    status held → expired
    active_held_quantity -= quantity
    append HOLD_EXPIRED
    store idempotent result:
        outcome = confirm_rejected_expired
    reject confirm
```

```
otherwise:
    Hold is still active
    continue
```

5. If this hold used extra capacity,
 check the coverage stored at Hold time
 against the resulting CONFIRMED disposition.

Net consumed may be unchanged.

That is not sufficient.
 Hold-only coverage → reject; the row stays held.
 Closed-to-new-use does not erase coverage that
 was already valid for confirmation.

6. Same row:
 status held → confirmed
 confirmed_at = now
7. Position aggregates only:
 active_held_quantity -= quantity
 confirmed_quantity += quantity
8. If confirmed_quantity > physical_capacity,
 escalate the commercial intervention case.
 Assert uncovered_overallocation = 0.
9. Outbox allocation.confirmed
 Store idempotent result

COMMIT

There are actually three possible committed outcomes, not simply success/failure:

```

Confirm command
├── Hold expired
│   ├── → materialise expiry
│   ├── → COMMIT
│   └── → return rejected-expired
├── Hold active but confirmation authority invalid
│   ├── → no allocation transition
│   ├── → store rejected result
│   └── → COMMIT
└── Hold active + valid
    ├── → held → confirmed
    ├── → update aggregates/risk/outbox
    ├── → COMMIT
    └── → return confirmed
  
```

Why the position lock stays

Suppose:

```

physical capacity = 6
held              = 2
  
```

```
confirmed      = 3
```

One transaction is confirming a Hold:

```
held          2 → 1
confirmed     3 → 4
```

while another trader is placing a new Hold and calculating:

```
bookable
=
physical
- held
- confirmed
```

Confirm does not take new stock, but it still writes held and confirmed on `inventory_position`. A concurrent Hold is reading that same pair to compute bookable. Without the row lock, one command can decide against a half-updated pair.

Confirm must validate **ownership, expiry, current state and resulting-disposition authority**, not merely the capacity arithmetic.

Race 3 — the transaction committed but the client never found out

Suppose:

```
database COMMIT succeeds
  |
  ▼
network response disappears
  |
  ▼
client retries
```

Without idempotency, that may create a second hold.

Every inventory-mutating command therefore requires:

```
idempotency key
+
```

```
request fingerprint
```

Example:

```
First request:
```

```
key          ABC
command      Hold 2
fingerprint X

COMMIT succeeds
```

The client retries:

```
key          ABC
command      Hold 2
fingerprint X
```

The server returns the stored successful result.

It does not create another hold.

If the same key arrives with a different fingerprint:

```
key          ABC
command      Hold 3
fingerprint Y
```

the server rejects it.

The goal is not exactly-once message delivery.

It is:

effectively-once business behaviour under retries.

Partial availability is a commercial decision

Suppose:

```
requested = 4
bookable  = 2
```

The system must not silently create:

```
Hold 2
```

Instead it returns a non-binding offer:

```
2 currently holdable  
2 unavailable
```

The trader can discuss this with the advertiser.

If the advertiser accepts:

```
Hold      2  
Pending   2
```

the trader submits that exact decision.

No database transaction remains open while the human conversation happens.

Race 4 — stale partial acceptance

Suppose the advertiser accepted:

```
Hold      2  
Pending   2
```

Before the trader submits that acceptance, another trader consumes one unit.

Current availability is now:

```
1
```

The system must not silently turn the advertiser's decision into:

```
Hold      1  
Pending   3
```

Instead:

```
allocation mutation = none

new non-binding offer:
  Holdable = 1
```

The trader must obtain a new commercial decision.

That protects both:

- inventory correctness;
- the meaning of the advertiser's consent.

Explicit commercial exception versus accidental over-allocation

The other part of the design I would be deliberately strict about is overselling.

Suppose:

```
Physical capacity = 6
```

The business may explicitly decide that it is willing to sell seven.

That can be a valid commercial decision.

But these two states must never become equivalent:

- ```
A. A human explicitly authorised +1 before allocation.

B. A race, migration or bug accidentally created booking #7.
```

I would never implement overselling as:

```
if user_is_privileged:
 ignore capacity
```

or:

```
allow negative availability
```

Instead there is a separate object:

```
extra_capacity_allowance

store
format
cycle

quantity

eligible actor / campaign where required
eligible disposition / action

reason
requested_by
approved_by
approved_at

status
```

Physical capacity remains: 6. It never become 7.

For a command that is eligible to use the allowance:

```
Physical capacity 6
Applicable approved extra 1
Effective limit for this command 7
```

The two quantities stay separate and auditable.

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## Authorised commercial risk

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Consider:

```
Physical capacity 6
Confirmed 6
Active holds 1

Physical overcapacity 1
Valid authorised coverage 1
Uncovered overallocation 0
```

That is:

**authorised commercial risk**

The allocation transaction should durably create or update a commercial intervention case.

The case belongs to the inventory position.

I would not arbitrarily label one advertiser as “the oversold advertiser”.

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## Accidental over-allocation

---

Now consider:

|                           |   |
|---------------------------|---|
| Physical capacity         | 6 |
| Confirmed                 | 6 |
| Active holds              | 1 |
| Physical overcapacity     | 1 |
| Valid authorised coverage | 0 |
| Uncovered overallocation  | 1 |

That is categorically different.

It is a:

**correctness / data-integrity incident**

The system should:

- preserve the evidence;
- block further ordinary consumption of that inventory key;
- investigate the cause;
- assess whether a permission/security bypass occurred;
- reconcile the state.

It should not silently convert that state into accepted oversell risk.

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## Prior authorisation matters

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This sequence must never rewrite history:

```
bug creates booking #7
 |
 ▼
someone notices
 |
 ▼
human creates +1 allowance
 |
```



▼  
system now labels booking #7 "authorised"

The allowance must have existed and been applicable **before the consuming command**.

A later allowance may authorise future actions.

It cannot retroactively legitimise an earlier defect.

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## Resulting disposition matters

---

Assume an allowance can authorise:

HOLD

but not:

CONFIRMATION

The system currently has:

|            |   |
|------------|---|
| Physical   | 6 |
| Confirmed  | 6 |
| Extra Hold | 1 |

The trader attempts:

HELD → CONFIRMED

A simplistic check might say:

total consumption changed by 0  
therefore confirmation is safe

That is insufficient.

The correct question is:

**Does valid exceptional authority exist for the resulting CONFIRMED disposition?**

This keeps:

temporary exceptional reservation

distinct from:

exceptional confirmed commitment

even when their capacity arithmetic is identical.

## Explicit allowance-policy assumption

For this slice I would model an allowance as explicitly declaring which dispositions it may cover:

HOLD  
CONFIRM  
or both

I would not assume all approvals automatically apply to both.

Likewise, closing an allowance to **new use** does not erase historical coverage already consumed while it was valid.

Whether an already-covered **Hold** may subsequently **Confirm** after closure depends on the allowance's confirmed-disposition policy.

That is a business rule, not something I would hide inside arithmetic.

## 3. Key trade-offs

| Choice                                              | Why I chose it                                                                                     | What could change later                                                                              |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Modular monolith over microservices</b>          | Allocation, allowance evidence, audit and outbox can commit atomically in one database transaction | Extract services when scaling, team ownership or operational boundaries justify the distributed cost |
| <b>PostgreSQL row locks over optimistic retries</b> | Contention is expected, concurrency is modest and the lock grain maps cleanly to the domain        | Benchmark OCC or <b>SERIALIZABLE</b> if measured lock contention becomes significant                 |
| <b>Fungible quantity over named physical units</b>  | The current product sells quantities such as <b>6 aisle barriers</b> , not                         | Add a fulfilment / position-assignment model if installation later                                   |

| Choice                                                | Why I chose it                                                       | What could change later                                                                             |
|-------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|                                                       | specific physical slots                                              | needs individual identities                                                                         |
| <b>Per-item outcomes over campaign-wide atomicity</b> | Supports mixed outcomes and keeps transactions narrow                | Introduce explicit atomic groups if future business rules require several items to succeed together |
| <b>Snapshot availability over reservation-on-read</b> | Avoids speculative reservations during browsing                      | Improve read freshness independently if measurements justify it                                     |
| <b>Transactional aggregate + allocation facts</b>     | Allocation decisions remain fast while history stays reconstructable | Revisit storage strategy only if measured volume or reporting requirements justify it               |

## What I consciously kept shallow

This is a depth choice rather than an omission.

### Trader workspace

I would expect:

- accessible bulk selection;
- virtualised large store lists;
- keyboard workflows;
- progressive batch results;
- visible hold expiry;
- conflict recovery;
- no optimistic display of an uncommitted Hold.

I have not gone deep on frontend state management because it does not determine whether inventory can be double-booked.

### Pricing and invoicing

Out of scope.

The important boundary is simply:

**Only confirmed allocations may appear downstream as confirmed commitments.**

### Installation

I am not designing field scheduling.

Accurate confirmed inventory and visibility of authorised physical overcapacity are integration constraints.

### Infrastructure topology

I would initially expect:

- stateless application instances;
- managed PostgreSQL;
- ordinary horizontal scaling;
- observability;
- backup and restore;
- CI/CD.

I see no requirement yet for microservice or Kubernetes complexity to solve the allocation problem.

## Commercial resolution of oversell

The system must detect and surface authorised physical risk.

The exact business process for resolving it remains a human workflow to define.

I would not silently invent cancellation or reassignment rules.

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## What would I do with more time, money or team?

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My first investment would be **proof and operability**, not more services.

I would build deterministic concurrency tests against the real PostgreSQL engine.

I would add property/model-based tests around quantity conservation and state transitions.

I would introduce failure injection around:

- network loss after commit;
- event publication failure;
- worker delay;
- transaction timeout;
- deadlock / retry;
- stale client actions.

I would improve reconciliation and commercial-risk tooling.

Only when measurements justified it would I consider:

- read replicas;
- CDC-fed availability projections;
- additional caches;
- service extraction;
- dedicated analytical infrastructure.

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## 4. How would I know it is working — and when it

# stops?

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A busy holds-per-hour graph is not evidence that the system is correct.

**Working means the invariants survive the races the domain creates.**

I would prove that first, then monitor it continuously.

---

## Tests I would run against real PostgreSQL

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### 1. Last physical unit

```
Two concurrent Hold(1)
physical available = 1
```

```
Expected:
exactly one succeeds
```

---

### 2. Last exceptional unit

```
Applicable extra allowance remaining = 1
```

```
Two concurrent exceptional Hold(1)
```

```
Expected:
exactly one succeeds
and exactly that allocation receives the coverage link
```

---

### 3. Stale partial acceptance

```
Offer:
Hold 2
Pending 2
```

```
Availability drops to 1 before acceptance
```

```
Expected:
no allocation mutation
revised non-binding offer returned
```

---

## 4. Confirmation at expiry

```
expires_at <= database_now
```

Confirm request arrives

Expected:  
confirmation rejected  
hold cannot be resurrected  
released capacity becomes ordinarily bookable  
pending demand receives nothing automatically

---

## 5. Allowance eligibility

Allowance valid for HOLD only

Exceptional Hold succeeds

Confirm attempted

Expected:  
Confirm fails unless confirmation-valid coverage exists

---

## 6. Unknown commit outcome

Hold commits  
HTTP response disappears  
client retries same idempotency key

Expected:  
stored original success  
exactly one Hold

---

## 7. Publisher unavailable

Confirmation commits  
outbox publisher unavailable

Expected:  
allocation remains confirmed

```
outbox row remains pending
publisher retries later
```

---

## 8. Reconciliation

```
SUM(active holds)
+
SUM(confirmed allocations)
```

must match:

```
inventory_position.active_held_quantity
+
inventory_position.confirmed_quantity
```

subject to the same authoritative expiry semantics.

---

## Continuous integrity signals

---

For each inventory key:

```
consumed
=
confirmed
+
active holds
```

Then:

```
physical_overcapacity
=
max(
 0,
 consumed - physical_capacity
)
```

and:

```
uncovered_overallocation
=
physical_overcapacity
-
valid_authorized_coverage
```

After every successful ordinary or authorised allocation command:

```
uncovered_overallocation = 0
```

That is the primary integrity invariant.

## Signal → likely meaning

| Signal                                                        | Likely meaning                                                                                             |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>uncovered_overallocation &gt; 0</b>                        | Defect, bad migration, unsafe administrative change or allocation path bypass                              |
| <b>Reconciliation drift ≠ 0</b>                               | Transactional aggregate and allocation facts disagree                                                      |
| <b>Confirm succeeds after expiry</b>                          | Time/expiry semantics differ between execution paths                                                       |
| <b>Partial acceptance succeeds with different quantity</b>    | System silently changed the recorded advertiser decision                                                   |
| <b>Physical exposure &gt; 0 but intervention case missing</b> | Authorised commercial risk has become operationally invisible                                              |
| <b>Confirmed outbox events ageing</b>                         | Source-of-record remains correct, but downstream commitment views are stale                                |
| <b>Lock wait / timeout rises materially</b>                   | Correctness may still hold, but contention is threatening availability and transaction design needs review |

This distinction matters:

```
capacity conflict
```

can be healthy.

If two traders request the final unit:



```
1 success
1 conflict
```

means the system worked.

By contrast:

```
high lock timeout
high deadlock retry
transaction timeout
```

is infrastructure contention.

I would not confuse scarce inventory with a broken database.

---

## Exceptional inventory signals

---

I would expose separately:

```
physical overcapacity

authorised oversell exposure

confirmed fulfilment shortfall

uncovered overallocation

open commercial intervention cases

open correctness incidents
```

For example:

|                   |   |
|-------------------|---|
| Physical capacity | 6 |
| Confirmed         | 7 |
| Valid coverage    | 1 |

means:

```
Uncovered overallocation 0
Confirmed fulfilment shortfall 1
```

The booking is authorised.

But the commercial risk has crystallised into more confirmed commitments than physical capacity.

That escalates the commercial intervention workflow.

The system should not silently cancel or amend the confirmation.

---

## Downstream health

---

Confirmation and its outbox event commit together.

Therefore:

```
DB COMMIT succeeds
event infrastructure fails
```

must result in:

```
confirmed allocation remains valid
outbox event remains pending
publication retries
```

I would monitor:

- oldest unpublished confirmed event;
- outbox backlog;
- publication retry rate;
- permanent publication failures;
- consumer deduplication failures.

The source of truth and downstream world may temporarily differ in freshness.

They must not differ in meaning.

---

## The dashboard I would want

---

## BOOKING HEALTH

---

### INVENTORY INTEGRITY

|                                 |   |
|---------------------------------|---|
| Uncovered overallocation        | 0 |
| Reconciliation drift            | 0 |
| Confirm-after-expiry violations | 0 |

### TRANSACTION HEALTH

Allocation attempts  
Successful allocations  
Expected business conflicts  
Lock wait  
Deadlock / transaction retry rate

### LIFECYCLE HEALTH

|                                  |   |
|----------------------------------|---|
| Active holds                     |   |
| Expired-hold materialisation lag |   |
| Idempotent replay failures       | 0 |

### COMMERCIAL RISK

Authorised physical overcapacity  
Confirmed fulfilment shortfall  
Open intervention cases

### INTEGRATION

|                                 |   |
|---------------------------------|---|
| Oldest unpublished confirmation |   |
| Permanent publication failures  | 0 |

I would establish concrete latency thresholds from representative tests rather than inventing them before seeing the deployment environment and normal workload.

The two figures I care about most are:

```
Uncovered overallocation = 0
Reconciliation drift = 0
```

because together they answer:

**Can I still trust what this system says has been reserved and confirmed?**

---

## Assumptions and open questions

---

I would keep these explicit rather than hiding them inside the design.

Proposed assumptions for this slice

- Physical units of one format within a store are fungible.
- A campaign is associated with one advertiser.
- The trader records advertiser decisions; no advertiser self-service workflow is assumed.
- Confirmation originates from an active Hold in the first slice.
- Authoritative time comes from the database.
- `expires_at <= now` means expired.
- PostgreSQL is the system of record.
- Extra-capacity allowances explicitly declare the dispositions they may authorise.

## Still open

- default and maximum Hold duration;
- Hold extension rules;
- exact approval/separation-of-duty policy for extra capacity;
- whether direct allocate-and-confirm is ever required;
- whether already-covered exceptional Holds may Confirm after allowance closure under each allowance policy;
- how a commercial intervention is eventually resolved;
- whether physical capacity may be corrected during an active trading period.

None of these prevents the core concurrency design from being implemented safely.

---

## AI tooling reflection

---

👉 See [AI tooling reflection](#)

## What I would build next

---

In order:

1. **inventory\_position + Hold / Release / Expire**  
Row locking, authoritative time, idempotency, audit and outbox.
2. **Confirm from an active Hold**  
Including expiry race and resulting-disposition validation.
3. **Extra-capacity allowance + coverage + case / incident**  
Prove that deliberate oversell and accidental over-allocation can never collapse into the same state.
4. **Campaign lines + partial offer / acceptance + pending remainder**  
Enforce quantity conservation and no silent shrink.
5. **Trader browse / read projections** Fast enough for thousands of stores. Still a snapshot.

That is the sequencing decision I would defend most strongly:

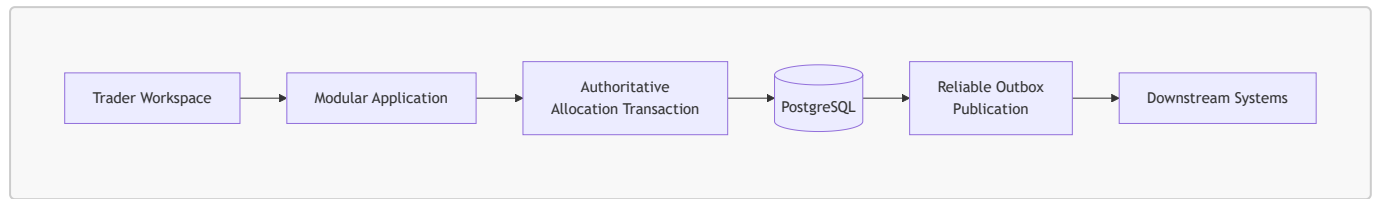
**Pin the write path before polishing the estate map.**

---

# Closing position

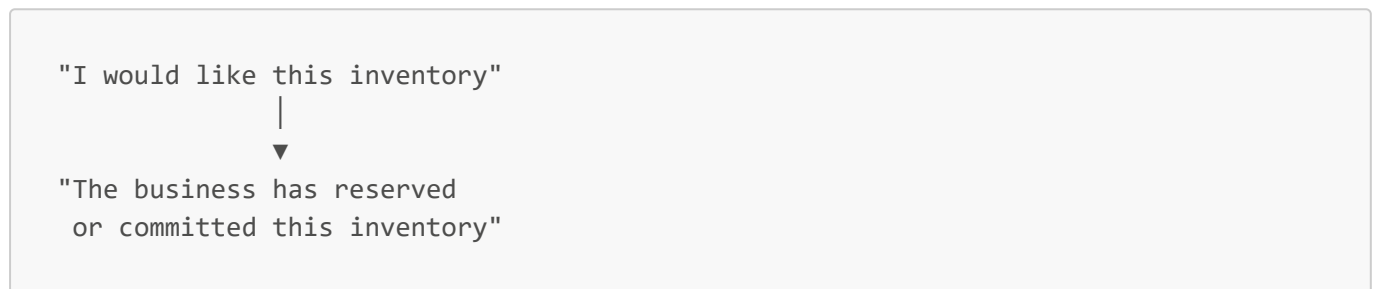
---

The overall architecture is deliberately conventional:



The sophistication belongs in the **correctness boundary**, not necessarily in the deployment topology.

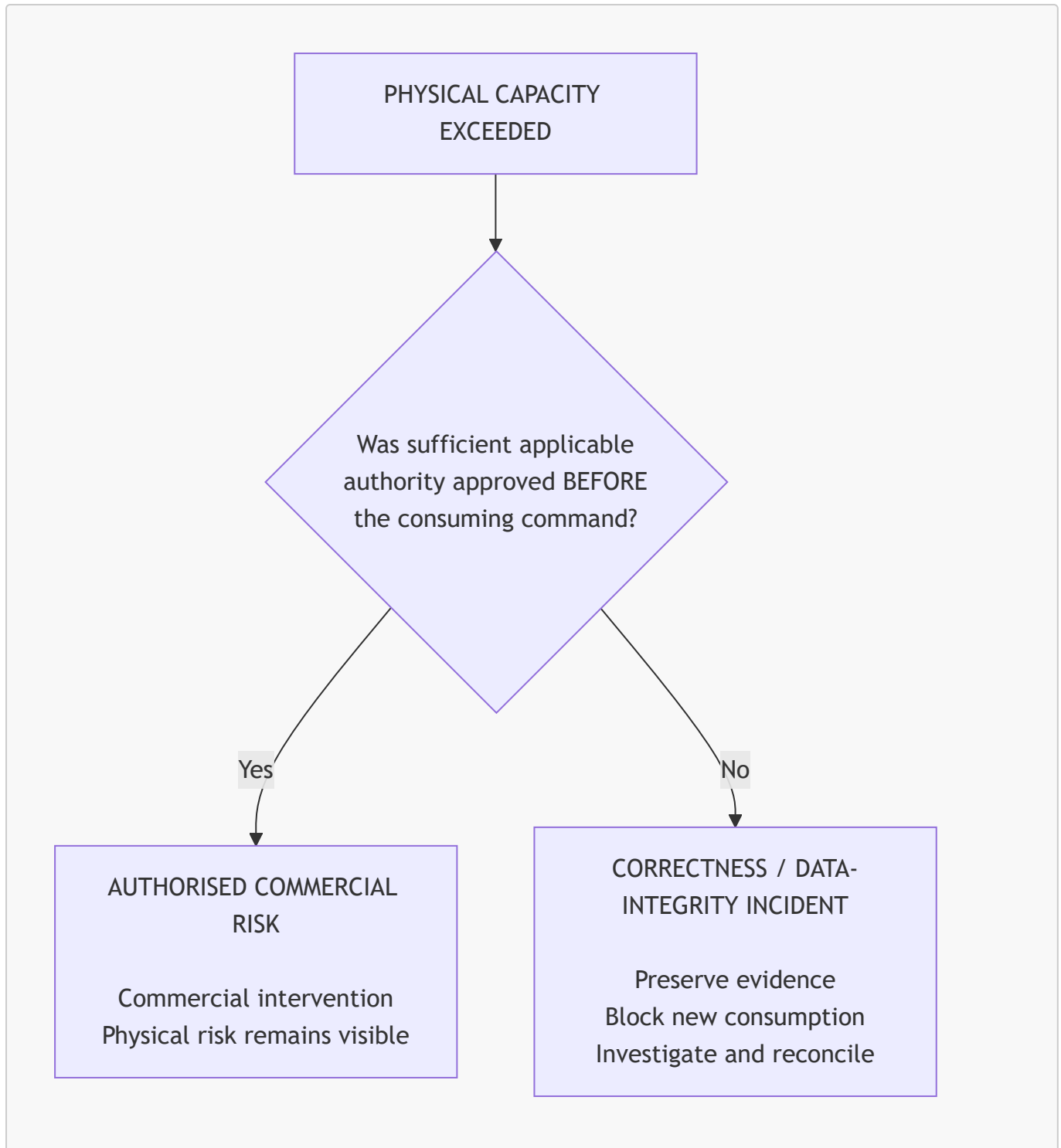
The transition I care about most is:



That transition must remain correct when:

- two traders act concurrently;
- availability is stale;
- holds expire;
- requests are retried after an unknown outcome;
- only part of the requested quantity is available;
- workers are delayed;
- downstream publication fails;
- the business deliberately authorises exceptional capacity.

And throughout all of those cases, one distinction must remain impossible to blur:



If that distinction survives contention, retries, expiry and failure, then the core booking lifecycle is doing its job.